QUBO-Inspired Portfolio Optimization with Sector and Budget Constraints

Quantum Dice Trinity Challenge 2026 — Stage 2 Submission

Abstract

We present a constraint-aware binary portfolio optimization framework inspired by Quadratic Unconstrained Binary Optimization (QUBO) principles. Binary asset-selection variables enable practical constraints—cardinality, sector diversification, and budget feasibility—to be encoded as quadratic penalty terms in a single energy function. The framework was developed incrementally across three versions (V1–V3), validated by exhaustive enumeration over $\binom{20}{6} = 38{,}760$ portfolios, and benchmarked against a classical Markowitz baseline. Historical return and covariance estimates for 20 S&P 100 stocks serve as inputs.

1 Introduction

Standard mean-variance optimization operates over continuous allocation weights and does not naturally accommodate discrete investment constraints—fixed portfolio size, sector exposure limits, or capital budgets. Casting asset selection as a binary combinatorial problem directly encodes these requirements and aligns with problem structures targeted by quantum and probabilistic optimization hardware. This work develops a QUBO-*inspired* energy formulation (not a fully explicit Q-matrix compilation) over a 20-stock universe spanning Technology, Finance, Healthcare, and Industrial/Energy sectors. The decision variable is $\mathbf{x} \in \{0,1\}^{20}$, where $x_i = 1$ indicates inclusion of asset i. Three incremental versions address cardinality (V1), sector diversification (V2), and budget feasibility (V3). Validation includes controlled stress tests and a classical Markowitz baseline. Historical daily closing prices for one year were sourced via `yfinance`; empirical expected returns $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$ were computed and exported to `mu.csv` and `Sigma.csv`.

2 Methodology

2.1 V1: Return-Risk with Cardinality

The base energy converts mean-variance optimization into a minimization over binary $\mathbf{x}$:

$$E_1(\mathbf{x}) = -\boldsymbol{\mu}^\top \mathbf{x} + \lambda\, \mathbf{x}^\top \boldsymbol{\Sigma} \mathbf{x} + \alpha\big(\textstyle\sum_i x_i - K\big)^2 \qquad (1)$$

The three terms represent expected return maximization, variance penalization (scaled by risk-aversion λ), and a hard cardinality constraint enforcing exactly $K = 6$ selected assets ($\alpha = 20$). Optimization proceeds by exhaustive enumeration of all $\binom{20}{6}$ feasible portfolios—tractable at this scale and guaranteeing global optima. A sweep over $\lambda \in \{0.1, 0.5, 1.0, 2.0\}$ confirmed that higher λ shifts composition toward lower-variance assets, as intended.

2.2 V2: Sector Diversification

V1 portfolios can concentrate entirely within one sector. V2 addresses this by penalizing sector over-exposure:

$$P_{\text{sector}}(\mathbf{x}) = \beta \sum_{j=1}^{4} \max\Big(0, \textstyle\sum_{i \in S_j} x_i - C_j\Big)^2 \qquad (2)$$

where S_j is the index set of sector j and $C_j = 2$ is the per-sector cap. In controlled testing with artificially elevated technology returns, $\beta = 0$ produced an all-technology four-stock portfolio. Setting $\beta = 10$ successfully enforced the sector cap in the controlled validation scenario, accepting a modest energy increase in exchange for diversification.

2.3 V3: Budget Feasibility

Diversified portfolios may still exceed affordable capital. V3 assigns representative prices $\mathbf{p}$ to each stock and introduces:

$$P_{\text{budget}}(\mathbf{x}) = \gamma\, \max\big(0,\, \mathbf{p}^\top \mathbf{x} - B\big)^2 \qquad (3)$$

with budget $B = \$1100$. A soft penalty formulation avoids infeasibility when budget, sector, and cardinality constraints interact.

2.4 Complete Energy Function

The full QUBO-inspired objective combines all terms:

$$\boxed{E(\mathbf{x}) = -\boldsymbol{\mu}^\top \mathbf{x} + \lambda\, \mathbf{x}^\top \boldsymbol{\Sigma} \mathbf{x} + \alpha\big(\textstyle\sum_i x_i - K\big)^2 + \beta\, P_{\text{sector}} + \gamma\, P_{\text{budget}}} \qquad (4)$$

V1 uses the first three terms; V2 adds βP_{sector}; V3 adds γP_{budget}.

3 Experimental Validation

3.1 Parameter Sweeps

A triple-nested sweep over $\lambda \in \{0.1, 0.5, 1.0, 2.0\}$, $\beta \in \{0, 0.1, 1, 10\}$, and $\gamma \in \{0, 0.01, 0.1, 1\}$ explored 64 parameter combinations in total. Increasing λ reduced portfolio variance at a marginal cost to expected return, confirming the intended risk–return tradeoff. Sector penalties activated only when diversification thresholds were exceeded, while budget penalties became binding only under controlled violations. This separation of effects improved interpretability of parameter tuning.

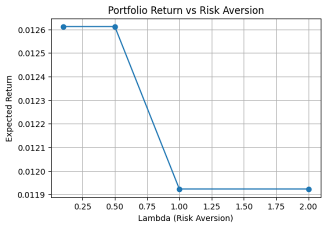

Figure 1: Expected return vs. λ.

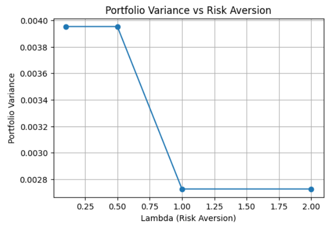

Figure 2: Portfolio variance vs. λ.

3.2 Budget Stress Test

A controlled experiment used six stocks with artificially elevated returns for the three most expensive and a tight budget of \$1200. With $\gamma = 0$, the optimizer selected {AAPL, MSFT, NVDA} at \$1800—a clear violation. With $\gamma = 1$, it shifted to {AAPL, MSFT, XOM} at exactly \$1200, retaining the two highest-return assets within budget. Across the full parameter sweep, 16 of 64 combinations produced violations at $\gamma = 0$; activating $\gamma \geq 0.01$ eliminated all violations. Further increasing γ did not materially change portfolio composition, indicating the budget level was meaningfully but not excessively binding.

4 Classical Baseline and Comparison

A Markowitz baseline was implemented via `scipy.optimize.minimize` (SLSQP), maximizing $\boldsymbol{\mu}^\top \mathbf{w} - \lambda\, \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}$ subject to $\sum_i w_i = 1$, $w_i \geq 0$. At $\lambda = 1.0$, the solution concentrated ∼59% in AMZN and ∼41% in MA—effectively two assets with no sector or budget constraints applied.

Table 1: Markowitz vs. V3 ($\lambda = 1.0$, $\beta = 1$, $\gamma = 0.01$).

Metric	Markowitz	V3
Expected Return	0.003782	0.011924
Variance	0.000282	0.002726
Asset Count	2	6
Representation	Continuous	Binary
Sector constraint	—	Satisfied
Budget constraint	—	Satisfied

The comparison highlights the tradeoff between unconstrained continuous allocation and constraint-aware binary selection. Markowitz achieves lower variance through fractional allocation, whereas the V3 formulation prioritizes practical investment constraints such as fixed cardinality, diversification, and affordability.

The comparison also reflects a representational distinction: Markowitz allocates fractional capital weights, whereas the proposed formulation selects discrete holdings through binary decision variables. While continuous allocation improves flexibility, binary selection naturally accommodates practical portfolio constraints.

5 Limitations and Future Work

- **Scalability.** Exhaustive enumeration over $\binom{n}{K}$ is feasible at $n = 20$ but scales combinatorially; larger universes require heuristic or hardware solvers.
- **Universe size.** The 20-stock universe is narrow relative to practical investment contexts.
- **Soft constraints.** Sector and budget penalties are soft; sufficiently large objective gains can outweigh them. Full enforcement requires auxiliary slack variables.
- **QUBO completeness.** The $\max(0, \cdot)$ inequality terms are not directly representable as a static Q-matrix. Explicit QUBO compilation with slack bits is deferred to future work and would enable direct hardware execution.
- **Solver scope.** Stage 2 prioritised formulation correctness and exact validation. Because exhaustive search over $\binom{20}{6} = 38{,}760$ portfolios remains tractable, it was preferred for rigour. Evaluation of probabilistic and annealing-based solvers is reserved for later stages involving larger search spaces and ORBIT-compatible workflows.
- **Penalty calibration.** Constraint performance depends on selecting appropriate penalty magnitudes (α, β, γ), which may require tuning for larger asset universes.

6 Conclusion

We developed and validated a QUBO-inspired binary portfolio optimization framework incorporating cardinality, sector, and budget constraints through an incremental V1→V2→V3 design. Controlled stress tests confirm that each penalty activates correctly and that modest γ values enforce budget feasibility without destabilizing the optimization. A Markowitz baseline contextualizes the tradeoffs introduced by binary selection. The formulation provides a transparent foundation for future work on explicit QUBO compilation, and Stage 3 experimentation will extend this to larger asset universes, probabilistic optimization methods, and ORBIT-compatible execution workflows. Beyond portfolio selection, the framework illustrates how practical investment constraints can be incrementally integrated into a unified optimization objective. This staged formulation process provides an interpretable pathway from classical optimization toward ORBIT-compatible experimentation in later stages.

References

[1] H. Markowitz, "Portfolio selection," *J. Finance*, vol. 7, pp. 77–91, 1952.

[2] F. Glover, G. Kochenberger, and Y. Du, "A tutorial on formulating and using QUBO models," *arXiv:1811.11538*, 2018.